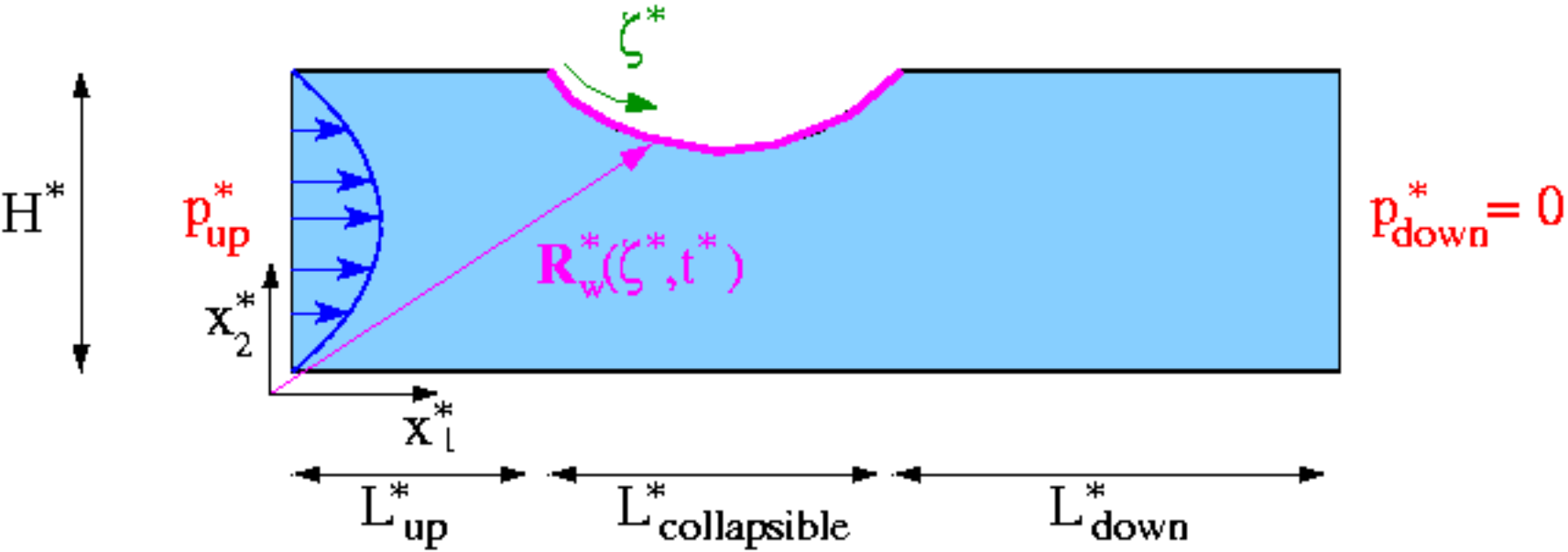
ζ^*
H^*
p^*_{up}
$p^*_{down} = 0$
$\mathbf{R}^*_w(\zeta^*, t^*)$
x^*_2
x^*_1
L^*_{up}
$L^*_{collapsible}$
L^*_{down}